

WIRELESS NETWORKING

List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

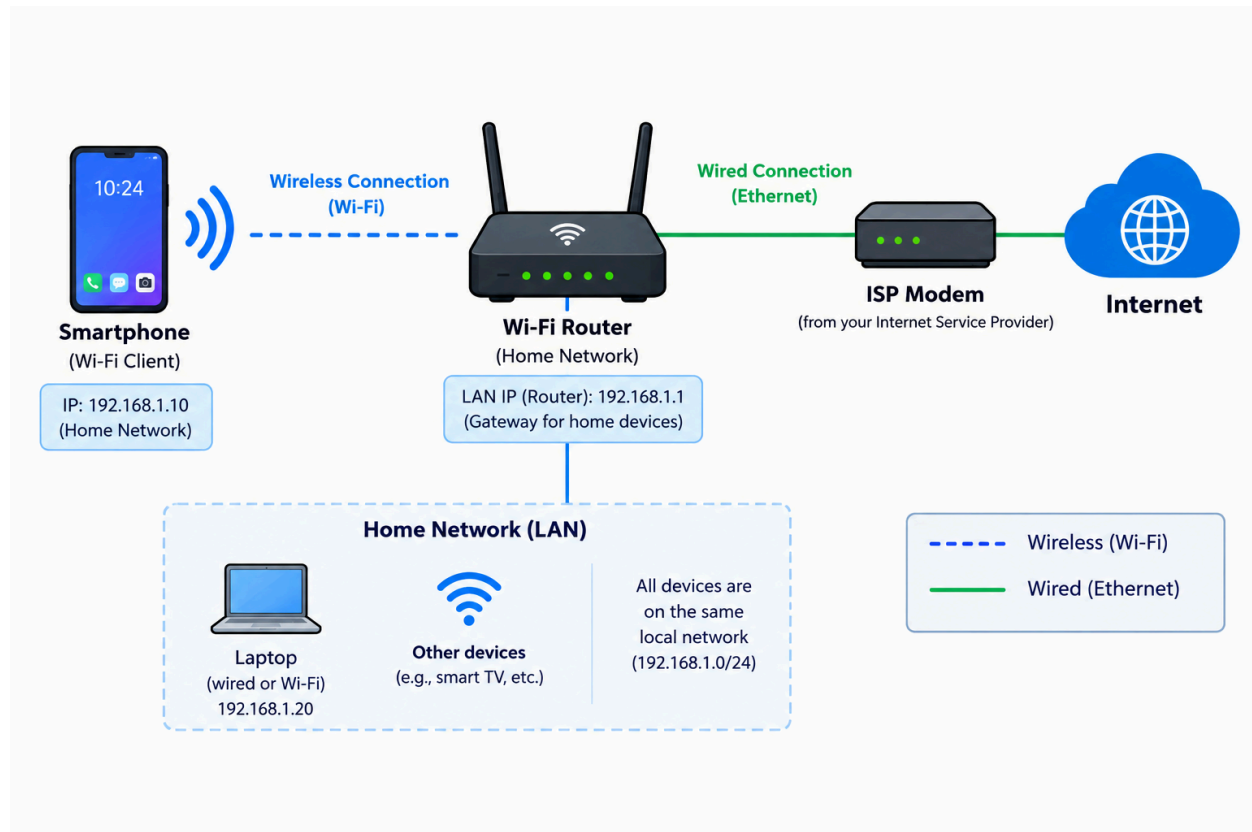
ANS:

Here are three popular Wi-Fi standards and their key differences in frequency and speed:

- **802.11n (Wi-Fi 4)**
 - **Key Difference:** Pioneered dual-band operation by supporting both the **2.4 GHz and 5 GHz bands**, achieving maximum theoretical speeds up to 600 Mbps using MIMO technology.
- **802.11ac (Wi-Fi 5)**
 - **Key Difference:** Operates **exclusively on the 5 GHz band** and introduces much wider channel bandwidths (up to 160 MHz), enabling multi-gigabit speeds up to 3.5 Gbps.
- **802.11ax (Wi-Fi 6)**
 - **Key Difference:** Expands operation across **2.4 GHz, 5 GHz, and the 6 GHz band** (Wi-Fi 6E), reaching theoretical speeds up to 9.6 Gbps with significantly better efficiency in crowded environments.

Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

ANS:



Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

ANS:

Wi-Fi Scan & Selection

- **Network A:** **Home_5G_Network** | **Signal:** Excellent (-45 dBm) | **Security:** WPA3-Personal
- **Network B:** **Neighbor_Extender** | **Signal:** Fair (-78 dBm) | **Security:** WPA2-PSK
- **Network C:** **Public_Airport_Guest** | **Signal:** Strong (-55 dBm) | **Security:** Open (No Encryption)

Choice & Rationale

Home_5G_Network is the clear choice. It combines an optimal signal strength for high throughput with **WPA3-Personal**, which utilizes the Simultaneous Authentication of Equals (SAE) handshake to protect against offline brute-force password cracking. In contrast, **Neighbor_Extender** suffers from a weak signal leading to packet loss, while

Public_Airport_Guest lacks encryption entirely, exposing all transmitted traffic to eavesdropping.

Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

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What is SSID Hiding?

The **SSID (Service Set Identifier)** is the visible name of your Wi-Fi network that appears in your device's dropdown list. **SSID hiding** (or disabling SSID broadcasting) configures your wireless router to stop advertising its name in beacon frames.

To connect to a hidden network, devices cannot simply select it from a list; a user must manually type the exact network name and security credentials.

Advantages and Disadvantages

- **Advantage: Reduced Visibility to Casual Observers**
 - *Why it helps:* It removes your network name from public view, preventing casual neighbors or passersby from seeing your network in their Wi-Fi drop-down lists and discouraging casual attempts to connect or guess passwords.
- **Disadvantage: False Sense of Security & Probing Leaks**
 - *Why it fails:* It does not provide real security. Your devices are constantly shouting out the hidden network's name into the air through **probe requests** as they look for the router. An attacker with a packet analyzer can easily intercept these probes and discover your hidden SSID in seconds. Furthermore, it causes connectivity inconvenience for legitimate guests.